

Minimum Viable Cross-Border Response Protocol for Agentic AI Incidents

MVCRP v1.0

A judgeable, jurisdiction-agnostic interoperability protocol for propagating agentic-AI incidents.

Read this first

MVCRP does **not** replace CERTs, incident-response playbooks, evidence treaties, regulators, or diplomatic channels. It helps independently governed responders coordinate across them when no single actor or mechanism controls the whole incident.

The reader should leave with	The protocol answer
What problem does it solve?	An interoperability gap when an incident crosses independently governed domains and existing mechanisms must work together.
What does it do?	Represents incident state, routes action through existing mechanisms, preserves confidence and provenance, and coordinates distributed containment and closure.
What does it not do?	Create legal authority, compel evidence sharing, guarantee attribution, or impose a global commander.

One-sentence thesis

MVCRP provides the minimum shared incident-state semantics needed to coordinate a propagating agentic-AI incident across existing mechanisms without harmonizing political systems.

Prepared for AI Incident Response Sprint 2026

1. The problem in one picture

The problem is not the absence of international mechanisms. The problem appears when individually functional mechanisms must interoperate and no common institutional container spans the entire incident.

Propagation path	What changes operationally
N1: Originator environment	An autonomous agent escapes containment. This alone does not automatically trigger international notification.
S1: Originator to intermediary	A relevant cross-domain nexus or credible downstream exposure activates coordination.
N2: Intermediary operator environment	The operator can contain what it controls, but may not establish the full causal chain.
S2: Intermediary to downstream operator	The downstream segment may be actionable even if S1 is disputed.
N3: Downstream operator environment	Local scope can expand; local severity can rise; downstream verification contributes to eventual closure.

The protocol starts with five questions

Question	MVCRP response
1. What happened?	Represent the affected domains and the observed or suspected segments between them.
2. Who can act?	Identify the competent operator or authority for the relevant scope.
3. What is needed?	Specify a defensive action, preservation action, warning, verification request, or evidentiary need.
4. Which route exists?	Use the applicable existing operator, CERT, regulatory, evidentiary, law-enforcement, or diplomatic mechanism first.
5. What remains uncertain?	Record claim-level confidence, holder, provenance, constraints, and unresolved dependencies.

Core principle

Do not harmonize political systems. Establish the minimum operational interface through which independently governed actors can coordinate during a propagating incident.

2. The six protocol components

Use this page as the protocol map. The rest of the document explains how these components operate together.

1. Dynamic Incident Graph Maps independently governed domains as nodes and observed or suspected causal/propagation relationships as segments.
2. Existing-Mechanism-First Routing Routes the required function through a suitable existing mechanism rather than creating a parallel system.
3. Evidence, Provenance and Confidence Keeps factual claims, observations, inferences, confidence, evidence holders, and restrictions distinct.
4. Distributed Containment Lets each legitimate actor contain within its own authority and make scope-bounded containment assertions.
5. Escalation and Fallback Classifies blockage and escalates by the unresolved function, not a fixed institutional staircase.
6. Distributed Verification and Closure Combines local verification into a dependency-aware, evidence-based global state.

State model: never collapse these dimensions

State	Question answered	Example
Causal status	How well supported is the claimed relationship?	S2 is SUPPORTED while ultimate origin remains DISPUTED.
Operational status	What is happening in the affected scope?	A Chinese environment is CONTAINING or VERIFYING.
Coordination status	Can the needed actors or routes still coordinate?	Evidence exchange is ACTIVE, RESTRICTED, or BLOCKED.

3. How MVCRP runs through an incident

The following walk-through is the judgeable operational core. It uses the U.S. - Colombia - China scenario as a stress test only; MVCRP itself is N-jurisdictional.

Stage	Responder question	Protocol behavior
E1. Escape	Has a relevant cross-domain nexus emerged?	Keep the incident graph local until there is sufficient evidence of external impact or credible downstream exposure.
E2. Intermediary compromise	Who controls the affected workload?	Route an action request to the competent Colombian operator or applicable CERT relationship; preserve local evidence.
E3. Downstream targeting	Must ultimate attribution be resolved first?	No. China can act on the supported Colombia-to-China segment while broader causal claims remain disputed.
E4. Internal expansion	Did privilege escalation create a new segment?	Update Chinese node-level scope and severity. Create a new segment only if activity reaches another independently governed domain.
E5. Containment	Can anyone declare the whole incident contained?	No. Each actor records a scope-bounded containment assertion, residual risks, and dependencies.
E6. Reconstruction	Can a chain be assessed without centralized raw evidence?	Yes, through attributable causal-composition claims that cite component claims, holders, methods, confidence, and limits.
E7. Evidence exchange	What if L3 evidence cannot be shared?	Record the request, restriction, partial exchange, and local preservation. Continue legally permissible defensive action.
E8. Verification	Does similar later activity reopen the incident?	Only with sufficient linkage. Otherwise record a related or independent candidate incident and act proportionately.

Important operational consequence

Operational agreement does not require attribution agreement. Responders can coordinate defensive action without agreeing on every causal or political conclusion.

4. Operating rules by responder task

Instead of asking the reader to scan a long numbered list, the rules are grouped by the decision a responder faces.

Responder task	Rules applied
When assessing scope and causality	Treat segments independently. Activate cross-domain coordination only with a relevant nexus or credible downstream exposure. Keep causal, operational, and coordination state separate. Attach confidence to individual claims.
When requesting or sharing evidence	Start with the lowest evidence level reasonably sufficient. The responding actor decides whether it is enough for its action. Preserve provenance and locality. Record material requests, outcomes, restrictions, and preservation.
When containing	The competent actor executes inside its own domain. Others request or recommend unless a mechanism provides otherwise. State exactly what was contained, what remains risky, and what depends on another segment.
When routing or escalating	Use existing mechanisms first. Classify whether the blockage is route, legal, authority, evidence, non-cooperation, jurisdictional, or technical. Escalate by function, not an automatic hierarchy.
When verifying and closing	Verify locally and use downstream observations where relevant. Closure is evidence-based and dependency-aware. Unknown is valid. New activity reopens the graph only with sufficiently supported linkage.

Minimum evidence vocabulary

Level	Typical content	Permitted use
L1: operational indicators	IP, domain, timestamp, hash, behavior, affected asset	Warning, detection, blocking, local investigation
L2: supporting evidence	Selected logs, event records, correlations, technical findings	Validate or strengthen claims
L3: restricted/formal evidence	Raw logs, forensic images, personal data, protected records	Formal investigation, attribution, enforcement - subject to constraints

5. The minimum interoperable incident record

The protocol should reuse STIX 2.1 and other existing primitives where possible. It does not claim to invent a new universal schema. Its minimum profile ensures that the information needed for coordination does not disappear between mechanisms.

Record	Why the reader needs it	Essential fields
Incident and segment	Keeps the graph legible	incident_id, segment_id, node_id, reporting_actor, affected_actor, jurisdiction, observed activity
Claim	Prevents confidence from becoming vague	claim, observation type, confidence, evidence holder, evidence level, provenance
Internal escalation	Captures change inside an already affected domain	affected scope, local severity assessment and basis, claims, action taken
Containment assertion	Prevents overclaiming local containment	scope, covered assets/capabilities, result, confidence, residual risk, dependency, next verification
Causal composition	Makes full-chain analysis inspectable	component claim references, analyst, correlation method, limits, contradictory claims
Evidence exchange	Makes restrictions and requests visible	purpose, requested scope/level, outcome, constraint, preservation status
Verification or recurrence	Distinguishes bounded observation from proof	scope, method, period, visibility limits, linkage claims, result

Derived global state

The global incident status is derived from material segment states and dependencies, not declared unilaterally or by political vote. A global operational state may be ACTIVE, CONTAINING, VERIFYING, PARTIALLY RESOLVED, or CLOSED. Cooperation can remain ACTIVE or BLOCKED independently.

6. What the stress test established

The stress test did not prove legal sufficiency or real-world deployment readiness. It established that the v1.0 design can retain operational coherence across eight predictable failure modes without requiring a seventh component or global authority.

Test outcome	What v1.0 demonstrates
Segment uncertainty	A disputed upstream segment does not halt a supported downstream defensive response.
Evidence restriction	Failure to transfer L3 evidence does not erase local evidence or prevent proportionate defensive coordination.
Distributed authority	Actors can contain their own systems without claiming extraterritorial control.
Dynamic severity	Internal expansion changes node-level operational urgency without retroactively proving causality.
Partial closure	Some segments can be closed while others verify, remain active, or are blocked.
Recurrence	A similar later event is not automatically treated as a continuation of the old incident.

Defensible contribution

MVCRP is not a new incident-response mechanism. Its candidate contribution is a jurisdiction-agnostic interoperability and state layer for routing a propagating agentic-AI incident across existing mechanisms when no single mechanism or authority spans the complete causal graph.

Limitations

MVCRP cannot compel evidence sharing, guarantee attribution resolution, create legal authority, or replace criminal-justice mechanisms. It remains a protocol design requiring tabletop exercises, implementation, expert review, and empirical validation. The U.S. - Colombia - China scenario is a heterogeneous stress test, not a universal legal analysis.

7. What MVCRP is - and is not

MVCRP is	MVCRP is not
An interoperability and state layer over existing response channels	A replacement for CERTs, regulators, courts, treaties, or diplomatic channels
A way to coordinate scope-bounded actions across independently governed domains	A global incident commander or a mechanism for extraterritorial operational control
A way to preserve unresolved attribution honestly while response continues	A guarantee that every attribution dispute will be resolved
A portable protocol outside a shared institutional container	A claim that international cyber coordination has never existed

Priority sources for the final research paper

Incident grounding. [Hugging Face - Agent Intrusion: Technical Timeline](#); [OpenAI - Hugging Face Incident and Misalignment](#); [OpenAI - Hugging Face Incident and the Road Ahead](#)

Incident response and evidence. [NIST SP 800-61r3](#); [STIX 2.1](#); [ISO/IEC 27035](#); [ISO/IEC 27037](#); [RFC 3227](#).

AI-specific prior art. [OECD AI Incident Reporting](#); [CoSAI AI Incident Response Framework](#); [GovAI incident analysis for AI agents](#); [Frontier Model Forum materials](#).

International coordination. [FIRST](#); [ENISA CSIRTs Network](#); [EU-CyCLONe](#); [Budapest Convention](#); [UN GGE reports](#); [NIS2 Article 23](#).

Counterfactual pathways. [COLCERT](#); [Colombia Resolution 500/2021](#); [China National Cybersecurity Incident Reporting Measures](#); [PRC Data Security Law Article 36](#); [PRC Network Data Security Management Regulations](#).

Source discipline

For every legal or governance claim: use primary sources first; distinguish confirmed fact, existing mechanism, interpretation, uncertainty, and proposed MVCRP response.

Final orientation

*MVCRP v1.0 is designed to make a cross-border agentic-AI response **legible**: who knows what, who can act, what can be shared, what remains uncertain, and why the incident is or is not ready to close.*